Doubles Strategy: The ENTER Duplicate Key

Doubles & Fact Strategies • Standard: CCSS.MATH.1.OA.C.6

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1. MATHEMATICAL CONCEPT & STACK APPROACH: Model doubles physically using matching digit blocks in Row 1 and Row 0 of the Place Value Tray and computationally using ENTER duplication.

Doubling a quantity means adding two identical copies of the same number (n + n = 2n). On a stack calculator, pressing [ENTER] duplicates the active number in X onto Waiting Shelf Y so pressing [+] immediately doubles it.

2. MODELING WITH BLOCKS (WHEN APPLICABLE) & STUDENT PREDICTION

Published Block Kit: Place Value Studio (Two-Row Tray & Digit Blocks)

Hands-On Setup: Place a [6] Digit Block in Row 1 (top) and a matching [6] Digit Block in Row 0 (bottom) of the Ones column on the Place Value Tray.

Student Prediction (Before Calculator): Observe the matching rows: 6 + 6 = 12 (1 Ten and 2 Ones on the tray).

3. CALCULATOR EXECUTION (SC-6 KEYCAPS & 4-LEVEL STACK TRACE)

Keystroke Flow: [6] [ENTER] [+]

Key Pressed	Screen (X)	Internal 4-Level Stack Effect (T, Z, Y, X)
[6]	6	Top row quantity (6) enters register X
[ENTER]	6	ENTER duplicates 6 into register Y! Both rows now on stack
[+]	12	Plus combines top and bottom rows: 6 + 6 = 12!

4. STORY CONTEXT (Twin Rows on the Workbench: How Matching Tray Rows Mirror Instant [ENTER] [+] Doubles)

When you place matching Digit Blocks in the top and bottom rows of your Place Value Tray, your eyes immediately recognize a double! In Chapters 1 & 2 of From Blocks to the Stack, pressing [ENTER] clones Workbench X onto Waiting Shelf Y just like matching Row 1 and Row 0 on the tray for instant [ENTER] [+] doubles.

Core Investigation (Student Warm-Up A):

Place [7] in both rows of your Place Value Tray. Use [ENTER] [+] to double it on the calculator.

Near-Doubles: Doubles Plus One

Fact Fluency & Near Doubles • Standard: CCSS.MATH.2.OA.B.2

1. EXTENSION CONCEPT (WARM-UP B), BLOCKS & PREDICTION

Concept & Stack Approach: A near-double (such as 7 + 8) is a double plus or minus 1 ((7 + 7) + 1). Evaluating the known double first leaves 14 on the stack, ready to combine with the +1 adjustment.

Blocks & Prediction: Place [7] in Row 1 (top) and [8] in Row 0 (bottom) of the Place Value Tray. Prediction: 7 and 8 is double 7 plus 1 more: 14 + 1 = 15.

2. EXTENSION CALCULATOR SEQUENCE & STACK TRACE

[7] [ENTER] [+] [1] [+]

Key Pressed	Screen (X)	Internal 4-Level Stack Effect
[7]	7	Enter base number 7
[ENTER]	7	Duplicate 7
[+]	14	Double: 7 + 7 = 14
[1]	1	Near-double difference (1)
[+]	15	14 + 1 = 15!

3. TARGET MISCONCEPTIONS & HANDS-ON BLOCK REMEDIATIONS

Misconception 1: Retyping the number to double it (e.g., [7] [ENTER] [7] [+]) instead of recognizing that [ENTER] duplicates Register X into Register Y.

Hands-On Block Remediation: Place matching [7] Digit Blocks in Row 1 and Row 0 of the Place Value Tray, then press [7] [ENTER] and point to the screen: both Y and X already hold 7, so pressing [+] doubles it immediately.

Misconception 2: Confusing '3 groups of 4' with '3 + 4' when transitioning from repeated addition to multiplication.

Hands-On Block Remediation: Model three groups of [4] on the Place Value Tray; have the student key [4] [ENTER] [4] [+] [4] [+] = 12 first, then compare directly with [4] [ENTER] [3] [×] = 12.

4. EXTENSION MATHEMATICAL INSIGHT

A near-double (such as 7 + 8) is a double plus or minus 1 ((7 + 7) + 1). Evaluating the known double first leaves 14 on the stack, ready to combine with the +1 adjustment.

Doubles, Near-Doubles & Equal Groups on the Place Value Tray

Name: _____ Date: _____ Class: _____

STUDENT REFERENCE GUIDE: TWIN-ROW DOUBLES & ENTER DUPLICATE RULES

- Placing matching Digit Blocks in Row 1 (top) and Row 0 (bottom) of the Place Value Tray models a physical Double.
- Doubles shortcut: Enter a number, press [ENTER] to duplicate it from Line X onto Line Y, then press [+]!
- Near-Doubles: Double the base number with [ENTER] [+], then press [1] [+] for one more.

Key Sequence: 6 ENTER +

Investigation 1: Calculate double 8 using [ENTER] [+].

1. Draw Your Blocks on the Tray:

2. Number Bond & Two Shelves:

Top Shelf (Line Y):

Bottom Shelf (Line X):

Combine (+ or −):

3. My Prediction:

Final Verified Answer:

[ ] Checked on SC-6

Investigation 2: Solve near-double 8 + 9: double 8 then add 1.

1. Draw Your Blocks on the Tray:

2. Number Bond & Two Shelves:

Top Shelf (Line Y):

Bottom Shelf (Line X):

Combine (+ or −):

3. My Prediction:

Final Verified Answer:

[ ] Checked on SC-6

Investigation 3: Calculate double 9 using [ENTER] [+].

1. Draw Your Blocks on the Tray:

2. Number Bond & Two Shelves:

Top Shelf (Line Y):

Bottom Shelf (Line X):

Combine (+ or −):

3. My Prediction:

Final Verified Answer:

[ ] Checked on SC-6

Investigation 4: Solve 5 + 6 using doubles plus one.

1. Draw Your Blocks on the Tray:

2. Number Bond & Two Shelves:

Top Shelf (Line Y):

Bottom Shelf (Line X):

Combine (+ or −):

3. My Prediction:

Final Verified Answer:

[ ] Checked on SC-6

Unit 04: Doubles, Near-Doubles & Equal Groups on the Place Value Tray — Answer Key

Complete solutions and SC-6 verification keystrokes for Grades 1–2

1. CORE (WARM-UP A) & EXTENSION (WARM-UP B) SOLUTIONS

Core Investigation (Warm-Up A): Place [7] in both rows of your Place Value Tray. Use [ENTER] [+] to double it on the calculator.

Core Solution: 14 ([1][4] on the tray). Keystrokes: [7] [ENTER] [+]

Core Check Keys: 7 ENTER +

Extension Challenge (Warm-Up B): Use the near-doubles strategy to solve 6 + 7 on the calculator.

Extension Solution: 13. Keystrokes: [6] [ENTER] [+] [1] [+]

Ext. Check Keys: 6 ENTER + 1 +

2. STUDENT LAB WORKSHEET SOLUTIONS (INVESTIGATIONS 1–4)

Investigation 1: Calculate double 8 using [ENTER] [+].

Expected Answer: 16

SC-6 Verification Keys: 8 ENTER +

Investigation 2: Solve near-double 8 + 9: double 8 then add 1.

Expected Answer: 17

SC-6 Verification Keys: 8 ENTER + 1 +

Investigation 3: Calculate double 9 using [ENTER] [+].

Expected Answer: 18

SC-6 Verification Keys: 9 ENTER +

Investigation 4: Solve 5 + 6 using doubles plus one.

Expected Answer: 11

SC-6 Verification Keys: 5 ENTER + 1 +

Hands-On Block & Stack Verification Protocol

Students first build the physical block model and write their mathematical strategy and prediction before verifying their result on the StackCalc calculator.